



CELEBAL

TECHNOLOGIES

Celebal Summer Internship 2026

Week 2 Task: Sample Superstore Dataset

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Objective

The objective of this assignment is to perform comprehensive sales data analysis using SQL by applying data exploration, filtering, aggregation, sorting, and business-oriented queries on the Superstore dataset. The goal is to extract meaningful insights regarding sales performance, customer behavior, product trends, and overall business profitability.

Methodology

The following steps were performed to complete the analysis:

1. Data Loading

- Imported the Superstore dataset into a MySQL database using Python and SQLAlchemy.
- Verified successful data insertion and row counts.

2. Data Exploration

- Examined the table schema and column structure.
- Retrieved sample records to understand the dataset.
- Validated the total number of records available for analysis.

3. Data Filtering

Applied SQL WHERE clauses to filter data based on:

- Region
- Product Category
- Order Date
- Sales Amount
- Multiple combined conditions

4. Aggregation Analysis

Used SQL aggregate functions and GROUP BY clauses to calculate:

- Total Sales
- Total Quantity Sold
- Average Sales
- Category-wise Sales and Profit

5. Sorting and Ranking

Performed sorting and ranking operations using ORDER BY and LIMIT to identify:

- Top-Selling Products
- Top Product Categories
- Most Profitable States

6. Business Use Cases

Solved real-world business problems including:

- Monthly Sales Trend Analysis
- Identification of Top Customers
- Customer Profitability Analysis
- Duplicate Record Detection

7. Data Validation

Conducted data quality checks by:

- Verifying total row counts
- Detecting duplicate records
- Checking for missing values
- Identifying negative-profit transactions

Data Loading Procedure

The Superstore dataset was imported into MySQL using Python, Pandas, and SQLAlchemy. The process involved loading the CSV dataset, converting date columns into DateTime format, creating the database automatically if it did not exist, establishing a connection with MySQL, loading the dataset into a table named **superstore**, and finally verifying the successful insertion of records.

Python Script Used for Data Loading

```
# Import required libraries

import pandas as pd

from sqlalchemy import create_engine, text

# MySQL Credentials

MYSQL_USER = "root"

MYSQL_PASSWORD = "YOUR_PASSWORD"

DATABASE_NAME = "superstore_db"
```

```

# Load CSV Dataset
df = pd.read_csv(
    "Sample - Superstore.csv",
    encoding="latin1"
)

# Convert Date Columns
df["Order Date"] = pd.to_datetime(
    df["Order Date"],
    errors="coerce"
)

df["Ship Date"] = pd.to_datetime(
    df["Ship Date"],
    errors="coerce"
)

# Create Database
server_engine = create_engine(
    f"mysql+pymysql://{MYSQL_USER}:{MYSQL_PASSWORD}@localhost"
)

with server_engine.connect() as conn:
    conn.execute(
        text(
            f"CREATE DATABASE IF NOT EXISTS {DATABASE_NAME}"
        )
    )
    conn.commit()

```

```

print(f"Database '{DATABASE_NAME}' is ready.")

# Connect Database

engine = create_engine(
    f"mysql+pymysql://{MYSQL_USER}:{MYSQL_PASSWORD}@localhost/{DATABASE_NAME}"
)

# Load Data into MySQL Table

df.to_sql(
    name="superstore",
    con=engine,
    if_exists="replace",
    index=False,
    chunksize=1000,
    method="multi"
)

print(f"Successfully inserted {len(df)} rows.")

# Verify Data Insertion

with engine.connect() as conn:

    result = conn.execute(
        text("SELECT COUNT(*) FROM superstore")
    )

    total_rows = result.scalar()

    print(f"Rows in MySQL Table: {total_rows}")

print("Data loading completed successfully.")

```

Output

```
Database 'superstore_db' is ready.  
Successfully inserted 9994 rows.  
Rows in MySQL Table: 9994  
Data loading completed successfully.
```

The output confirms that the database was created successfully, the Superstore dataset was loaded into MySQL, and all 9,994 records were inserted and verified successfully.

Data Exploration

The data exploration phase was performed to understand the structure and contents of the dataset before conducting further analysis.

Query 1: Examine Table Schema

The following query was executed to inspect the table structure and column information.

```
DESCRIBE superstore;
```

Output

	Field	Type	Null	Key	Default	Extra
►	Row ID	bigint	YES		NULL	
	Order ID	text	YES		NULL	
	Order Date	datetime	YES		NULL	
	Ship Date	datetime	YES		NULL	
	Ship Mode	text	YES		NULL	
	Customer ID	text	YES		NULL	
	Customer Name	text	YES		NULL	
	Segment	text	YES		NULL	
	Country	text	YES		NULL	
	City	text	YES		NULL	
	State	text	YES		NULL	
	Postal Code	bigint	YES		NULL	
	Region	text	YES		NULL	
	Product ID	text	YES		NULL	
	Category	text	YES		NULL	
	Sub-Category	text	YES		NULL	
	Product Name	text	YES		NULL	
	Sales	double	YES		NULL	
	Quantity	bigint	YES		NULL	
	Discount	double	YES		NULL	
	Profit	double	YES		NULL	

Observation

The table contains 21 columns with appropriate data types, including customer, product, sales, quantity, discount, profit, and order-related information required for analysis.

Query 2: Retrieve Sample Records

The following query was executed to view sample records from the dataset.

```
SELECT *
```

```
FROM superstore
```

```
LIMIT 10;
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
1	CA-2016-152156	2016-11-08 00:00:00	2016-11-11 00:00:00	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	Furniture	Bookcases	Bush Somerset Collection B
2	CA-2016-152156	2016-11-08 00:00:00	2016-11-11 00:00:00	Second Class	CG-12520	Claire Gule	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	Furniture	Chairs	Hon Deluxe Fabric Upholste
3	CA-2016-138688	2016-06-12 00:00:00	2016-06-16 00:00:00	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address Label
4	US-2015-108966	2015-10-11 00:00:00	2015-10-18 00:00:00	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series Slim
5	US-2015-108966	2015-10-11 00:00:00	2015-10-18 00:00:00	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	Office Supplies	Storage	Eldon Fold N Roll Cart Syst
6	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood an
7	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
8	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP p
9	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders w/it
10	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206/VEL 6 Outlet

Observation

Sample records confirm that the dataset has been loaded correctly.

Query 3: Validate Total Number of Records

The following query was executed to verify the total number of records.

```
SELECT COUNT (*) AS Total_Rows
```

```
FROM superstore;
```

Output

	Total_Rows
▶	9994

Observation

The dataset contains 9,994 records and is ready for analysis.

Data Filtering

SQL WHERE clauses were used to filter records based on region, product category, order date, sales amount, and multiple business conditions.

Query 4: Region Filter

```
SELECT *
```

```
FROM superstore
```

```
WHERE Region = 'West';
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
3	CA-2016-138688	2016-06-12 00:00:00	2016-06-16 00:00:00	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	Office Supplies	Labels	Self-Adhesive Address L
6	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	Furniture	Furnishings	Eldon Expressions Wood
7	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	Office Supplies	Art	Newell 322
8	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP
9	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	Office Supplies	Binders	DXL Angle-View Binders
10	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-10002892	Office Supplies	Appliances	Belkin F5C206/VEL 6 Ou
11	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular
12	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Korffel 250 Conference
14	CA-2016-161389	2016-12-05 00:00:00	2016-12-10 00:00:00	Standard Class	JA-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	Office Supplies	Binders	Fellowes PB200 Plastic C
18	CA-2014-167164	2014-05-13 00:00:00	2014-05-15 00:00:00	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	Office Supplies	Storage	Fellowes Super Stor/Dra
19	CA-2014-143336	2014-08-27 00:00:00	2014-09-01 00:00:00	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	Office Supplies	Art	Newell 341
20	CA-2014-143336	2014-08-27 00:00:00	2014-09-01 00:00:00	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phon
21	CA-2014-143336	2014-08-27 00:00:00	2014-09-01 00:00:00	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	Office Supplies	Binders	Wilson Jones Hanging Vi
25	CA-2015-106320	2015-09-25 00:00:00	2015-09-30 00:00:00	Standard Class	ES-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR4500 Series
26	CA-2016-121755	2016-01-16 00:00:00	2016-01-20 00:00:00	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001624	Office Supplies	Binders	Wilson Jones Active Use
27	CA-2016-121755	2016-01-16 00:00:00	2016-01-20 00:00:00	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini TravelC
43	CA-2016-101343	2016-07-17 00:00:00	2016-07-22 00:00:00	Standard Class	RA-19885	Ruben Ausman	Corporate	United States	Los Angeles	California	90049	West	OFF-ST-10003479	Office Supplies	Storage	Eldon Base for stackable
63	CA-2015-135545	2015-11-24 00:00:00	2015-11-30 00:00:00	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC-10004633	Technology	Accessories	Verbatim 25 GB 6x Blu-ra
64	CA-2015-135545	2015-11-24 00:00:00	2015-11-30 00:00:00	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OFF-BI-10001078	Office Supplies	Binders	Acco PRESTEXT Data Bin
65	CA-2015-135545	2015-11-24 00:00:00	2015-11-30 00:00:00	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	OFF-PA-10003892	Office Supplies	Paper	Xerox 1943
66	CA-2015-135545	2015-11-24 00:00:00	2015-11-30 00:00:00	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	FUR-FU-10000397	Furniture	Furnishings	Luxo Economy Swing Arr
68	CA-2014-106376	2014-12-05 00:00:00	2014-12-10 00:00:00	Standard Class	BS-11590	Brendan Sived	Corporate	United States	Gilbert	Arizona	85234	West	OFF-AR-10002871	Office Supplies	Art	Hunt BOSTON Model 16C

Observation

Records belonging to the West region were retrieved successfully.

Query 5: Product Category Filter

```
SELECT *
FROM superstore
WHERE Category = 'Technology';
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
8	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002275	Technology	Phones	Mitel 5320 IP Phone VoIP
12	CA-2014-115812	2014-06-09 00:00:00	2014-06-14 00:00:00	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TEC-PH-10002033	Technology	Phones	Konitel 250 Conference
20	CA-2014-140336	2014-08-27 00:00:00	2014-09-01 00:00:00	Second Class	ZD-21925	Zacharias Donastell	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10001949	Technology	Phones	Cisco SPA 501G IP Phone
27	CA-2016-121755	2016-01-16 00:00:00	2016-01-20 00:00:00	Second Class	BH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10003027	Technology	Accessories	Imation 8GB Mini Travel
36	CA-2016-117590	2016-12-08 00:00:00	2016-12-10 00:00:00	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4
41	CA-2015-117415	2015-12-27 00:00:00	2015-12-31 00:00:00	Standard Class	SN-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	TEC-PH-10000486	Technology	Phones	Plantronics HL10 Handse
42	CA-2017-120999	2017-09-10 00:00:00	2017-09-15 00:00:00	Standard Class	LC-16930	Linda Cazamas	Corporate	United States	Naperville	Illinois	60540	Central	TEC-PH-10000493	Technology	Phones	Panasonic Kx-TS555
45	CA-2016-118255	2016-03-11 00:00:00	2016-03-13 00:00:00	First Class	ON-18715	Odella Nelson	Corporate	United States	Eagan	Minnesota	55122	Central	TEC-AC-10000171	Technology	Accessories	Verbatim 25 GB 6x Blu-ray
48	CA-2016-169194	2016-06-20 00:00:00	2016-06-25 00:00:00	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-AC-10002167	Technology	Accessories	Imation 8gb Micro Travel
49	CA-2016-169194	2016-06-20 00:00:00	2016-06-25 00:00:00	Standard Class	LH-16900	Lena Hernandez	Consumer	United States	Dover	Delaware	19901	East	TEC-PH-10003988	Technology	Phones	HP Elite 30 Dazzle Design
55	CA-2016-105816	2016-12-11 00:00:00	2016-12-17 00:00:00	Standard Class	JM-15265	Janet Molinar	Corporate	United States	New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones	AT&T CLS451 4-Handse
60	CA-2016-111682	2016-06-17 00:00:00	2016-06-18 00:00:00	First Class	TB-21055	Ted Butterfield	Consumer	United States	Troy	New York	12180	East	TEC-AC-10002167	Technology	Accessories	Imation 8gb Micro Travel
63	CA-2015-135545	2015-11-24 00:00:00	2015-11-30 00:00:00	Standard Class	KM-16720	Kunst Miller	Consumer	United States	Los Angeles	California	90004	West	TEC-AC-10004633	Technology	Accessories	Verbatim 25 GB 6x Blu-ray
69	CA-2014-106376	2014-12-05 00:00:00	2014-12-10 00:00:00	Standard Class	BG-11590	Brendan Siveed	Corporate	United States	Gilbert	Arizona	85234	West	TEC-PH-10002726	Technology	Phones	netTALK DUO VoIP Tele
87	CA-2017-155358	2017-10-26 00:00:00	2017-11-02 00:00:00	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	TEC-AC-10001998	Technology	Accessories	Logitech LS21 Speaker S
91	CA-2016-109806	2016-09-17 00:00:00	2016-09-22 00:00:00	Standard Class	JS-15685	Jim Sink	Corporate	United States	Los Angeles	California	90036	West	TEC-PH-10004093	Technology	Phones	Panasonic Kx-TS550
101	CA-2016-198568	2016-08-29 00:00:00	2016-09-02 00:00:00	Standard Class	RB-19465	Rick Bersley	Home Office	United States	Chicago	Illinois	60610	Central	TEC-AC-10001767	Technology	Accessories	SanDisk Ultra 64 GB Micr
104	US-2015-156867	2015-11-13 00:00:00	2015-11-17 00:00:00	Standard Class	LC-16970	Lena Cacoppo	Consumer	United States	Aurora	Colorado	80013	West	TEC-AC-10001552	Technology	Accessories	Logitech C350 2.4Ghz W
107	CA-2017-119004	2017-11-23 00:00:00	2017-11-28 00:00:00	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Ca...	28205	South	TEC-AC-10003499	Technology	Accessories	Memorex Mini Travel On
108	CA-2017-119004	2017-11-23 00:00:00	2017-11-28 00:00:00	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Ca...	28205	South	TEC-PH-10002844	Technology	Phones	Speck Products Candysh
110	CA-2015-129476	2015-10-15 00:00:00	2015-10-20 00:00:00	Standard Class	PA-19060	Pete Armstrong	Home Office	United States	Orland Park	Illinois	60462	Central	TEC-AC-10000844	Technology	Accessories	Logitech Gaming G510s
124	CA-2016-103730	2016-06-12 00:00:00	2016-06-15 00:00:00	First Class	SC-20725	Steven Cartwright	Consumer	United States	Wilmington	Delaware	19805	East	TEC-PH-10003875	Technology	Phones	KLD Oscar II Style Snap

Observation

Technology category products were isolated for analysis.

Query 6: Order Date Filter

```
SELECT *
FROM superstore
WHERE `Order Date`
BETWEEN '2017-01-01'
AND '2017-12-31';
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
13	CA-2017-114412	2017-04-15 00:00:00	2017-04-20 00:00:00	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	Office Supplies	Paper	Xerox 1967
24	US-2017-156909	2017-07-16 00:00:00	2017-07-18 00:00:00	Second Class	SP-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FLR-CH-10002774	Furniture	Chairs	Global Deluxe Stad
35	CA-2017-107727	2017-10-19 00:00:00	2017-10-23 00:00:00	Second Class	MA-17560	Matt Abelman	Home Office	United States	Houston	Texas	77095	Central	OFF-PA-10000249	Office Supplies	Paper	Easy-staple paper
42	CA-2017-120999	2017-09-10 00:00:00	2017-09-15 00:00:00	Standard Class	LC-16930	Linda Cazamas	Corporate	United States	Naperville	Illinois	60540	Central	TEC-PH-10000493	Technology	Phones	Panasonic Kx-TS555
44	CA-2017-139619	2017-09-19 00:00:00	2017-09-23 00:00:00	Standard Class	ES-14080	Erin Smith	Corporate	United States	Melbourne	Florida	32935	South	OFF-ST-10003282	Office Supplies	Storage	Advantus 10-Draw
72	CA-2017-114440	2017-09-14 00:00:00	2017-09-17 00:00:00	Second Class	TB-21520	Tracy Blumstein	Consumer	United States	Jackson	Michigan	49201	Central	OFF-PA-10004675	Office Supplies	Paper	Telephone Message
76	US-2017-118038	2017-12-09 00:00:00	2017-12-11 00:00:00	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OFF-BI-10004182	Office Supplies	Binders	Economy Binders
77	US-2017-118038	2017-12-09 00:00:00	2017-12-11 00:00:00	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	FLR-FU-10000260	Furniture	Furnishings	6" Cubicle Wall Clo
78	US-2017-118038	2017-12-09 00:00:00	2017-12-11 00:00:00	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OFF-ST-10000615	Office Supplies	Storage	SimpliFile Personal f
85	US-2017-119662	2017-11-13 00:00:00	2017-11-16 00:00:00	First Class	CS-12400	Christopher Schild	Home Office	United States	Chicago	Illinois	60623	Central	OFF-ST-10003656	Office Supplies	Storage	Safco Industrial W
86	CA-2017-140088	2017-05-28 00:00:00	2017-11-16 00:00:00	Standard Class	PO-18865	Patrick O'Donnell	Consumer	United States	Columbia	South Carolina	29203	South	FLR-CH-10000863	Furniture	Chairs	Novimex Swivel Fal
87	CA-2017-155358	2017-10-26 00:00:00	2017-11-02 00:00:00	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	TEC-AC-10001998	Technology	Accessories	Logitech LS21 Spee
88	CA-2017-155358	2017-10-26 00:00:00	2017-11-02 00:00:00	Standard Class	PG-18895	Paul Gonzalez	Consumer	United States	Rochester	Minnesota	55901	Central	OFF-PA-10000134	Office Supplies	Labels	Avery 511
96	US-2017-109484	2017-11-06 00:00:00	2017-11-12 00:00:00	Standard Class	RB-19705	Roger Barrio	Home Office	United States	Portland	Oregon	97206	West	OFF-BI-10004738	Office Supplies	Binders	FlexiFile Leather-Lc
97	CA-2017-161018	2017-11-09 00:00:00	2017-11-11 00:00:00	Second Class	PH-18775	Parthena Horris	Home Office	United States	New York...	New York	10009	East	FLR-FU-10000629	Furniture	Furnishings	9-3/4 Diameter Ro
98	CA-2017-159833	2017-06-17 00:00:00	2017-06-20 00:00:00	First Class	KD-16345	Katherine Douch	Consumer	United States	San Franci...	California	94122	West	OFF-BI-10001721	Office Supplies	Binders	Trimflex Flexible Po
107	CA-2017-119004	2017-11-23 00:00:00	2017-11-28 00:00:00	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Carolina	28205	South	TEC-AC-10003499	Technology	Accessories	Memorex Mini Trav
108	CA-2017-119004	2017-11-23 00:00:00	2017-11-28 00:00:00	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Carolina	28205	South	TEC-PH-10002844	Technology	Phones	Speck Products Car
109	CA-2017-119004	2017-11-23 00:00:00	2017-11-28 00:00:00	Standard Class	JM-15250	Janet Martin	Consumer	United States	Charlotte	North Carolina	28205	South	OFF-AR-10000390	Office Supplies	Art	Novell Chalk Hold
111	CA-2017-146780	2017-12-25 00:00:00	2017-12-30 00:00:00	Standard Class	CV-12805	Cynthia Voltz	Corporate	United States	New York...	New York	10035	East	FLR-FU-10001934	Furniture	Furnishings	Magnifier Swing Ar
127	US-2017-107272	2017-11-05 00:00:00	2017-11-12 00:00:00	Standard Class	TS-21610	Troy Staebe	Consumer	United States	Phoenix	Arizona	85023	West	OFF-BI-10003174	Office Supplies	Binders	Avery Durable Slen
128	US-2017-107272	2017-11-05 00:00:00	2017-11-12 00:00:00	Standard Class	TS-21610	Troy Staebe	Consumer	United States	Phoenix	Arizona	85023	West	OFF-ST-10002974	Office Supplies	Storage	Travel-File Heavy-I

Observation

Records from the specified date range were retrieved successfully.

Query 7: Sales Amount Filter

*SELECT **

FROM superstore

WHERE Sales > 1000;

Output

	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
▶	Standard Class	BH-11710	Brosna Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	Furniture	Tables	Chromcraft Rectangular Conference Tables	1706.184	9	0.2	85.3092
	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	Furniture	Tables	Bretford CR-6000 Series Slim Rectangular Table	1044.63	3	0	240.2649
	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palais Royal Lawyers Bookcase, Royal...	3083.43	7	0.5	-1665.0522
	First Class	GH-14485	Gene Hale	Corporate	United States	New York City	Texas	75080	Central	TEC-PH-10004977	Technology	Phones	GE 30524EE4	1097.544	7	0.2	123.4737
	Standard Class	JM-15265	Janet Molinari	Corporate	United States	New York City	New York	10024	East	TEC-PH-10002447	Technology	Phones	AT&T CL83451 4-Handset Telephone	1029.95	5	0	298.6855
	Standard Class	BS-11590	Brendan Sweed	Corporate	United States	Gilbert	Arizona	85234	West	OFF-AR-10002671	Office Supplies	Art	Hunt BOSTON Model 1606 High-Volume Electric ...	1113.024	8	0.2	111.3024
	Standard Class	JE-16165	Justin Ellison	Corporate	United States	Franklin	Wisconsin	53132	Central	FUR-CH-10004054	Furniture	Chairs	Hon Deluxe Fabric Upholstered Stacking Chairs,...	1951.84	8	0	585.552
	Standard Class	BH-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	TEC-MA-10000822	Technology	Machines	Lexmark M6611dhe Monochrome Laser Printer	8159.952	8	0.4	-1359.992
	Standard Class	BH-11140	Becky Martin	Consumer	United States	San Antonio	Texas	78207	Central	FUR-CH-10004287	Furniture	Chairs	SAFCO Arcus Folding Chair	1740.06	9	0.3	-24.858
	Standard Class	VW-21275	Victoria Wilson	Corporate	United States	Medina	Ohio	44256	East	TEC-MA-10000864	Technology	Machines	Cisco 9971 IP Video Phone Charcoal	1188	9	0.7	-950.4
	Second Class	DW-13480	Dianna Wilson	Home Off...	United States	Lakeville	Minnesota	55044	Central	FUR-CH-10004063	Furniture	Chairs	Global Deluxe High-Back Manager's Chair	2001.86	7	0	580.5394
	Second Class	DW-13480	Dianna Wilson	Home Off...	United States	Lakeville	Minnesota	55044	Central	OFF-AP-10002945	Office Supplies	Appliances	Honeywell Enviraclear Portable HEPA Air Cleane...	1503.25	5	0	496.0725
	Standard Class	KC-16540	Kelly Colister	Consumer	United States	San Diego	California	92037	West	TEC-AC-10003832	Technology	Accessories	Logitech P710e Mobile Speakerphone	3347.37	13	0	636.0003
	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central	TEC-MA-10000822	Technology	Machines	Lexmark M6611dhe Monochrome Laser Printer	3059.982	3	0.4	-509.997
	Second Class	SB-20290	Sean Braxton	Corporate	United States	Houston	Texas	77036	Central	TEC-MA-10001353	Technology	Machines	Xerox WorkCentre 6505DN Laser Multifunction ...	2519.958	7	0.4	-251.9958
	Second Class	SC-20725	Steven Cartwright	Consumer	United States	Houston	Texas	77070	Central	TEC-PH-10001760	Technology	Phones	Bose SoundLink Bluetooth Speaker	1114.4	7	0.2	376.11
	Standard Class	JO-15280	Jas O'Carroll	Consumer	United States	Los Angeles	California	90004	West	FUR-TA-10002903	Furniture	Tables	Bevis Round Bulfinch 29" High Table Top	1038.84	5	0.2	51.942
	Standard Class	JO-15895	Jonathan Doherty	Corporate	United States	Belleville	New Jersey	7109	East	OFF-ST-10002756	Office Supplies	Storage	Tenneco Star-O-Stor Boltless Shelving, 5 Shelve...	1082.48	8	0	10.8248
	Standard Class	NM-18445	Nathan Mautz	Home Off...	United States	New York City	New York	10024	East	TEC-MA-10002927	Technology	Machines	Canon imageCLASS MF7460 Monochrome Digital...	3991.98	2	0	1995.99
	First Class	AM-10060	Adam Bellavance	Home Off...	United States	New York City	New York	10009	East	OFF-BI-10004995	Office Supplies	Binders	GBC DocoBind P400 Electric Binding System	4355.168	4	0.2	1415.4296
	Jame Day	MK-17905	Michael Kennedy	Consumer	United States	Manchester	Connecticut	6040	East	FUR-BO-10002545	Furniture	Bookcases	Atlantic Metals Mobile 3-Shelf Bookcases, Custo...	1043.92	4	0	271.4192
	Standard Class	AG-10900	Arthur Gainer	Consumer	United States	Tucson	Arizona	85705	West	TEC-AC-10003911	Technology	Accessories	NETGEAR AC1750 Dual Band Gigabit Smart WiFi R...	1023.936	8	0.2	176.1888
													NETGEAR AC1750 Dual Band Gigabit Smart WiFi Router				

Observation

High-value sales transactions were identified.

Query 8: Combined Filter

*SELECT **

FROM superstore

WHERE Region='West'

AND Category='Technology'

AND Sales > 1000;

Output

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
▶	252	CA-2016-145625	2016-09-11 00:00:00	2016-09-17 00:00:00	Standard Class	KC-16540	Kelly Colister	Consumer	United States	San Diego	California	92037	West	TEC-AC-10003832	Technology	Accessories	Logitech P710e Mobile S
	374	US-2014-119137	2014-07-23 00:00:00	2014-07-27 00:00:00	Standard Class	AG-10900	Arthur Gainer	Consumer	United States	Tucson	Arizona	85705	West	TEC-AC-10003911	Technology	Accessories	NETGEAR AC1750 Dual
	393	US-2014-135972	2014-09-21 00:00:00	2014-09-23 00:00:00	Second Class	JG-15115	Jack Garza	Consumer	United States	Des Moines	Washington	98198	West	TEC-CO-10002313	Technology	Copiers	Canon PC1080F Person
	407	CA-2017-117457	2017-12-08 00:00:00	2017-12-12 00:00:00	Standard Class	KH-16510	Keith Herrera	Consumer	United States	San Francisco	California	94110	West	TEC-CO-10004115	Technology	Copiers	Sharp AL-1530CS Digita
	516	CA-2017-127432	2017-01-22 00:00:00	2017-01-27 00:00:00	Standard Class	AD-10180	Alan Dominguez	Home Office	United States	Great Falls	Montana	59405	West	TEC-CO-10003236	Technology	Copiers	Canon Image Class D66
	596	CA-2014-131450	2014-08-08 00:00:00	2014-08-15 00:00:00	Standard Class	LR-16915	Lena Radford	Consumer	United States	San Diego	California	92024	West	TEC-CO-10004115	Technology	Copiers	Sharp AL-1530CS Digita
	754	CA-2016-110499	2016-04-07 00:00:00	2016-04-09 00:00:00	First Class	YC-21895	Yoseph Carroll	Corporate	United States	San Francisco	California	94110	West	TEC-CO-10002095	Technology	Copiers	Hewlett Packard 610 Co
	807	CA-2014-134313	2014-11-01 00:00:00	2014-11-07 00:00:00	Standard Class	RM-16915	Russell Applegate	Consumer	United States	Denver	Colorado	80219	West	TEC-PH-10001795	Technology	Phones	ClearOne CHATAttach
	815	CA-2017-143686	2017-05-14 00:00:00	2017-05-14 00:00:00	Same Day	Standard Class	Pauline Johnson	Consumer	United States	Santa Ana	California	92704	West	TEC-AC-10001838	Technology	Accessories	Razer Tiamat Over Ear
	1049	CA-2014-139192	2014-05-27 00:00:00	2014-06-01 00:00:00	Second Class	LC-16885	Lena Creighton	Consumer	United States	San Francisco	California	94109	West	TEC-PH-10000486	Technology	Phones	Plantronics HL10 Hands
	1487	CA-2015-140410	2015-11-03 00:00:00	2015-11-07 00:00:00	Standard Class	CM-12655	Corinna Mitchell	Home Office	United States	Los Angeles	California	90008	West	TEC-PH-10003580	Technology	Phones	Cisco IP Phone 7961G-C
	1626	CA-2015-157084	2015-12-19 00:00:00	2015-12-24 00:00:00	Standard Class	JG-15160	James Galang	Consumer	United States	Los Angeles	California	90049	West	TEC-AC-10002049	Technology	Accessories	Plantronics Savi W720P
	1645	CA-2015-111820	2015-03-19 00:00:00	2015-03-20 00:00:00	First Class	PH-14365	Fred Hopkins	Corporate	United States	Seattle	Washington	98115	West	TEC-CO-10001766	Technology	Copiers	Canon PC340 Copier
	1806	US-2016-116729	2016-12-25 00:00:00	2016-12-28 00:00:00	First Class	GK-14620	Grace Kelly	Corporate	United States	Los Angeles	California	90049	West	TEC-PH-10002200	Technology	Phones	Samsung Galaxy Note 2
	2114	CA-2015-164882	2015-10-31 00:00:00	2015-10-31 00:00:00	Same Day	SG-20080	Sandra Glasco	Consumer	United States	Redlands	California	92374	West	TEC-PH-10001578	Technology	Phones	Polycom SoundStation2
	2221	CA-2017-130841	2017-07-28 00:00:00	2017-08-01 00:00:00	Standard Class	MH-17620	Matt Hagelstein	Corporate	United States	San Francisco	California	94110	West	TEC-PH-10001817	Technology	Phones	Wilson Electronics DB Ph
	2274	CA-2015-142944	2015-03-06 00:00:00	2015-03-11 00:00:00	Standard Class	JL-15850	John Lucas	Consumer	United States	San Francisco	California	94122	West	TEC-CO-10003763	Technology	Copiers	Canon PC1060 Persona
	2405	CA-2017-108574	2017-10-07 00:00:00	2017-10-11 00:00:00	Standard Class	MG-18145	Mike Gockenbach	Consumer	United States	Los Angeles	California	90045	West	TEC-AC-10002049	Technology	Accessories	Logitech G19 Programm
	2419	CA-2016-107202	2016-05-21 00:00:00	2016-05-28 00:00:00	Standard Class	LC-16930	Linda Cazamas	Corporate	United States	Sparks	Nevada	89431	West	TEC-MA-10000112	Technology	Machines	Panasonic KX MB2061P
	2549	US-2015-128090	2015-08-16 00:00:00	2015-08-22 00:00:00	Standard Class	JM-15865	John Murray	Consumer	United States	Phoenix	Arizona	85023	West	TEC-PH-10002624	Technology	Phones	Samsung Galaxy S4 Min
	2710	CA-2015-131797	2015-01-30 00:00:00	2015-02-06 00:00:00	Standard Class	CC-12145	Charles Crestani	Consumer	United States	Los Angeles	California	90049	West	TEC-MA-10000488	Technology	Machines	Baby BDC10 JPRU Card
	2794	CA-2014-154599	2014-04-12 00:00:00	2014-04-17 00:00:00	Standard Class	KH-16450	Kean Nguyen	Corporate	United States	Redondo Beach	California	90278	West	TEC-PH-10001557	Technology	Phones	Pyle PMP37LED

Observation

Records matching all specified business conditions were retrieved.

Aggregation Analysis

Aggregation analysis was performed using SQL aggregate functions and GROUP BY clauses to summarize the dataset and evaluate key business metrics.

Query 9: Total Sales by Region

```
SELECT  
    Region,  
    ROUND(SUM(Sales),2) AS Total_Sales  
FROM superstore  
GROUP BY Region;
```

Output

	Region	Total_Sales
▶	South	391721.91
	West	725457.82
	Central	501239.89
	East	678781.24

Observation

Total sales were calculated for each region.

Query 10: Total Quantity Sold by Category

```
SELECT  
    Category,  
    SUM(Quantity) AS Total_Quantity  
FROM superstore  
GROUP BY Category;
```

Output

	Category	Total_Quantity
▶	Furniture	8028
	Office Supplies	22906
	Technology	6939

Observation

The total quantity sold was determined for each product category.

Query 11: Average Sales by Category

SELECT

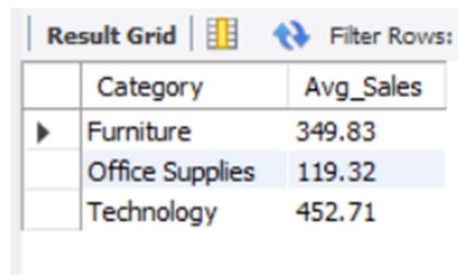
Category,

ROUND(AVG(Sales),2) AS Avg_Sales

FROM superstore

GROUP BY Category;

Output



The screenshot shows a 'Result Grid' with a 'Filter Rows' button. The grid contains three rows of data. The first row is the header with columns 'Category' and 'Avg_Sales'. The second row shows 'Furniture' with an average sales of 349.83. The third row shows 'Office Supplies' with an average sales of 119.32. The fourth row shows 'Technology' with an average sales of 452.71.

	Category	Avg_Sales
▶	Furniture	349.83
	Office Supplies	119.32
	Technology	452.71

Observation

Average sales values were calculated for each category.

Query 12: Category-wise Sales and Profit

SELECT

Category,

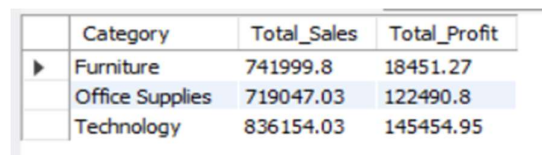
ROUND(SUM(Sales),2) AS Total_Sales,

ROUND(SUM(Profit),2) AS Total_Profit

FROM superstore

GROUP BY Category;

Output



The screenshot shows a table with four columns: 'Category', 'Total_Sales', and 'Total_Profit'. There are three rows of data. The first row is the header. The second row shows 'Furniture' with total sales of 741999.8 and total profit of 18451.27. The third row shows 'Office Supplies' with total sales of 719047.03 and total profit of 122490.8. The fourth row shows 'Technology' with total sales of 836154.03 and total profit of 145454.95.

	Category	Total_Sales	Total_Profit
▶	Furniture	741999.8	18451.27
	Office Supplies	719047.03	122490.8
	Technology	836154.03	145454.95

Observation

Category-wise sales and profit performance were analyzed.

Sorting and Ranking

Sorting and ranking operations were performed using ORDER BY and LIMIT clauses to identify top-performing products, categories, and states based on sales and profit metrics.

Query 13: Top 10 Products by Sales

```
SELECT
    `Product Name`,
    ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY `Product Name`
ORDER BY Total_Sales DESC
LIMIT 10;
```

Output

	Product Name	Total_Sales
►	Canon imageCLASS 2200 Advanced Copier	61599.82
	Fellowes PB500 Electric Punch Plastic Comb Bind...	27453.38
	Cisco TelePresence System EX90 Videoconferen...	22638.48
	HON 5400 Series Task Chairs for Big and Tall	21870.58
	GBC DocuBind TL300 Electric Binding System	19823.48
	GBC Ibimaster 500 Manual ProClick Binding System	19024.5
	Hewlett Packard LaserJet 3310 Copier	18839.69
	HP Designjet T520 Inkjet Large Format Printer -...	18374.9
	GBC DocuBind P400 Electric Binding System	17965.07
	High Speed Automatic Electric Letter Opener	17030.31

Observation

The highest revenue-generating products were identified.

Query 14: Top Product Categories by Sales

```
SELECT
    Category,
    ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY Category
```

ORDER BY Total_Sales DESC;

Output

	Category	Total_Sales
▶	Technology	836154.03
	Furniture	741999.8
	Office Supplies	719047.03

Observation

Product categories were ranked based on total sales performance.

Query 15: Most Profitable States

SELECT

State,

ROUND(SUM(Profit),2) AS Total_Profit

FROM superstore

GROUP BY State

ORDER BY Total_Profit DESC

LIMIT 10;

Output

	State	Total_Profit
▶	California	76381.39
	New York	74038.55
	Washington	33402.65
	Michigan	24463.19
	Virginia	18597.95
	Indiana	18382.94
	Georgia	16250.04
	Kentucky	11199.7
	Minnesota	10823.19
	Delaware	9977.37

Observation

The most profitable states were identified based on total profit generated.

Business Use Cases

Business-oriented SQL queries were executed to analyze sales trends, identify valuable customers, evaluate customer profitability, and validate data quality through duplicate record detection.

Query 16: Monthly Sales Trend Analysis

```
SELECT
    YEAR(`Order Date`) AS Year,
    MONTH(`Order Date`) AS Month,
    ROUND(SUM(Sales),2) AS Monthly_Sales
FROM superstore
GROUP BY YEAR(`Order Date`),
    MONTH(`Order Date`)
ORDER BY Year, Month;
```

Output

	Year	Month	Monthly_Sales
▶	2014	1	14236.89
	2014	2	4519.89
	2014	3	55691.01
	2014	4	28295.34
	2014	5	23648.29
	2014	6	34595.13
	2014	7	33946.39
	2014	8	27909.47
	2014	9	81777.35
	2014	10	31453.39
	2014	11	78628.72
	2014	12	69545.62
	2015	1	18174.08
	2015	2	11951.41
	2015	3	38726.25
	2015	4	34195.21
	2015	5	30131.69
	2015	6	24797.29
	2015	7	28765.32
	2015	8	36898.33

Observation

Monthly sales trends were analyzed to understand revenue patterns over time.

Query 17: Top Customers by Sales

```
SELECT
    `Customer Name`,
    ROUND(SUM(Sales),2) AS Total_Sales
FROM superstore
GROUP BY `Customer Name`
ORDER BY Total_Sales DESC
LIMIT 10;
```

Output

	Customer Name	Total_Sales
▶	Sean Miller	25043.05
	Tamara Chand	19052.22
	Raymond Buch	15117.34
	Tom Ashbrook	14595.62
	Adrian Barton	14473.57
	Ken Lonsdale	14175.23
	Sanjit Chand	14142.33
	Hunter Lopez	12873.3
	Sanjit Engle	12209.44
	Christopher Conant	12129.07

Observation

Top revenue-contributing customers were identified.

Query 18: Customer Profitability Analysis

```
SELECT
    `Customer Name`,
    ROUND(SUM(Profit),2) AS Total_Profit
FROM superstore
GROUP BY `Customer Name`
ORDER BY Total_Profit DESC
LIMIT 10;
```

Output

	Customer Name	Total_Profit
▶	Tamara Chand	8981.32
	Raymond Buch	6976.1
	Sanjit Chand	5757.41
	Hunter Lopez	5622.43
	Adrian Barton	5444.81
	Tom Ashbrook	4703.79
	Christopher Martinez	3899.89
	Keith Dawkins	3038.63
	Andy Reiter	2884.62
	Daniel Raglin	2869.08

Observation

The most profitable customers were identified based on total profit generated.

Query 19: Duplicate Record Detection

SELECT

`Order ID`,

COUNT () AS Duplicate_Count*

FROM superstore

GROUP BY `Order ID`

HAVING COUNT () > 1;*

Output

	Order ID	Duplicate_Count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	7
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4
	CA-2014-106376	2
	US-2015-134026	3
	US-2017-118038	3
	CA-2016-127208	2
	CA-2014-139451	2
	CA-2017-155558	2
	CA-2016-109806	3
	CA-2015-149587	3
	CA-2016-158568	3
	US-2015-156867	3
	CA-2017-119004	2

Observation

Duplicate Order IDs were identified for data validation and transaction analysis.

Data Validation

Data quality checks were performed to verify data completeness, consistency, and reliability before drawing business insights.

Query 20: Verify Total Row Count

```
SELECT COUNT (*) AS Total_Rows
```

```
FROM superstore;
```

Output

	Total_Rows
▶	9994

Observation

The dataset contains 9,994 records, confirming successful data loading.

Query 21: Detect Duplicate Records

```
SELECT
```

```
    `Order ID`,
```

```
    COUNT (*) AS Duplicate_Count
```

```
FROM superstore
```

```
GROUP BY `Order ID`
```

```
HAVING COUNT (*) > 1;
```

Output

	Order ID	Duplicate_Count
▶	CA-2016-152156	2
	US-2015-108966	2
	CA-2014-115812	7
	US-2015-118983	2
	CA-2014-143336	3
	CA-2016-137330	2
	CA-2016-121755	2
	US-2015-150630	7
	CA-2016-117590	2
	CA-2015-117415	4
	CA-2016-118255	2
	CA-2016-169194	2
	CA-2015-115742	4
	CA-2016-105816	2
	CA-2016-111682	7
	CA-2015-135545	4
	CA-2014-106376	2
	US-2015-134026	3
	US-2017-118038	2

Observation

Duplicate Order IDs were identified for validation purposes.

Query 22: Check for Missing Values

```
SELECT *  
  
FROM superstore  
  
WHERE Sales IS NULL;
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name	Sales	Quantity	Discount	Profit
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Observation

The dataset was checked for missing values in critical fields.

Query 23: Identify Negative-Profit Transactions

```
SELECT *  
  
FROM superstore  
  
WHERE Profit < 0;
```

Output

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Category	Sub-Category	Product Name
4	US-2015-108966	2015-10-11 00:00:00	2015-10-18 00:00:00	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	Furniture	Tables	Bretford CR4500
15	US-2015-118983	2015-11-22 00:00:00	2015-11-26 00:00:00	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	Office Supplies	Appliances	Holmes Replacem
16	US-2015-118983	2015-11-22 00:00:00	2015-11-26 00:00:00	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	Office Supplies	Binders	Stonex DuraTech
24	US-2017-156909	2017-07-16 00:00:00	2017-07-18 00:00:00	Second Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10002774	Furniture	Chairs	Global Deluxe Sta
28	US-2015-150630	2015-09-17 00:00:00	2015-09-21 00:00:00	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	Furniture	Bookcases	Riverside Palas R
29	US-2015-150630	2015-09-17 00:00:00	2015-09-21 00:00:00	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474	Office Supplies	Binders	Avery Recycled F
33	US-2015-150630	2015-09-17 00:00:00	2015-09-21 00:00:00	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10001525	Office Supplies	Binders	Acco Pressboard
37	CA-2016-117590	2016-12-08 00:00:00	2016-12-10 00:00:00	First Class	GH-14485	Gene Hale	Corporate	United States	Richardson	Texas	75080	Central	FUR-FU-10003664	Furniture	Furnishings	Electrix Architect
39	CA-2015-117415	2015-12-27 00:00:00	2015-12-31 00:00:00	Standard Class	SH-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	FUR-BO-10002545	Furniture	Bookcases	Atlantic Metals M
40	CA-2015-117415	2015-12-27 00:00:00	2015-12-31 00:00:00	Standard Class	SH-20710	Steve Nguyen	Home Office	United States	Houston	Texas	77041	Central	FUR-CH-10004218	Furniture	Chairs	Global Fabric Mar
67	US-2015-164175	2015-04-30 00:00:00	2015-05-05 00:00:00	Standard Class	PS-18970	Paul Stevenson	Home Office	United States	Chicago	Illinois	60610	Central	FUR-CH-10001146	Furniture	Chairs	Global Value Mid-I
73	US-2015-134026	2015-04-26 00:00:00	2015-05-02 00:00:00	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South	FUR-CH-10000513	Furniture	Chairs	High-Back Leather
75	US-2015-134026	2015-04-26 00:00:00	2015-05-02 00:00:00	Standard Class	JE-15745	Joel Eaton	Consumer	United States	Memphis	Tennessee	38109	South	OFF-BI-10004123	Office Supplies	Storage	Safo Industrial I
76	US-2017-118038	2017-12-09 00:00:00	2017-12-11 00:00:00	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	OFF-BI-10004182	Office Supplies	Binders	Economy Binders
77	US-2017-118038	2017-12-09 00:00:00	2017-12-11 00:00:00	First Class	KB-16600	Ken Brennan	Corporate	United States	Houston	Texas	77041	Central	FUR-FU-10000260	Furniture	Furnishings	6" Cubicle Wall C
79	US-2014-147606	2014-11-26 00:00:00	2014-12-01 00:00:00	Second Class	JE-15745	Joel Eaton	Consumer	United States	Houston	Texas	77070	Central	FUR-FU-10003194	Furniture	Furnishings	Eldon Expression
85	US-2014-147662	2017-11-13 00:00:00	2017-11-16 00:00:00	First Class	CS-12400	Christopher Schild	Home Office	United States	Chicago	Illinois	60623	Central	OFF-ST-10003656	Office Supplies	Storage	Safo Industrial I
96	US-2017-109484	2017-11-06 00:00:00	2017-11-12 00:00:00	Standard Class	RB-19705	Roger Barcio	Home Office	United States	Portland	Oregon	97206	West	OFF-BI-10004738	Office Supplies	Binders	Flexible Leather-
101	CA-2016-158568	2016-08-29 00:00:00	2016-09-02 00:00:00	Standard Class	RB-19465	Rick Benley	Home Office	United States	Chicago	Illinois	60610	Central	TEC-AC-10001767	Technology	Accessories	SanDisk Ultra 64
102	CA-2016-158568	2016-08-29 00:00:00	2016-09-02 00:00:00	Standard Class	RB-19465	Rick Benley	Home Office	United States	Chicago	Illinois	60610	Central	OFF-BI-10002609	Office Supplies	Binders	Avery Hidden Tal
104	US-2015-156867	2015-11-13 00:00:00	2015-11-17 00:00:00	Standard Class	LC-16870	Lena Cacoppo	Consumer	United States	Aurora	Colorado	80013	West	TEC-AC-10001552	Technology	Accessories	Logitech K350 2-
105	US-2015-156867	2015-11-13 00:00:00	2015-11-17 00:00:00	Standard Class	LC-16870	Lena Cacoppo	Consumer	United States	Aurora	Colorado	80013	West	FUR-FU-10004906	Furniture	Furnishings	Deflect-o-DuraMa
106	US-2015-156867	2015-11-13 00:00:00	2015-11-17 00:00:00	Standard Class	LC-16870	Lena Cacoppo	Consumer	United States	Aurora	Colorado	80013	West	OFF-BI-10002794	Office Supplies	Binders	Avery Trapezoid
119	US-2015-136476	2015-04-05 00:00:00	2015-04-10 00:00:00	Standard Class	GG-14650	Greg Guthrie	Corporate	United States	Bristol	Tennessee	37620	South	OFF-BI-10003650	Office Supplies	Binders	GBC DoolBind 30
125	US-2014-152030	2014-12-26 00:00:00	2014-12-28 00:00:00	Second Class	AD-10180	Alan Dominguez	Home Office	United States	Houston	Texas	77041	Central	FUR-CH-10004063	Furniture	Chairs	Global Deluxe Ho
152	US-2016-134514	2016-09-30 00:00:00	2016-09-26 00:00:00	Standard Class	DE-10165	Robin Fox	Consumer	United States	Bloomington	Texas	61701	Central	OFF-TA-10004034	Office Supplies	Tables	Bretford CR4500

Observation

Loss-making transactions were identified for profitability analysis.

Brief Insights

Based on the SQL analysis performed on the Sample Superstore Dataset; the following key insights were derived:

1. The dataset contains 9,994 sales records and was successfully loaded into the MySQL database.
2. The schema includes customer, product, sales, quantity, discount, profit, and regional attributes required for comprehensive business analysis.
3. Data filtering enabled targeted analysis based on region, category, sales amount, and date ranges.
4. Aggregation analysis provided summarized business metrics such as total sales, total quantity sold, average sales, and category-wise performance.
5. Sorting and ranking operations identified the top-selling products, best-performing categories, and most profitable states.
6. Monthly sales trend analysis revealed variations in revenue across different periods, helping understand sales patterns over time.
7. Customer analysis identified high-value customers who contributed significantly to overall sales and profit.
8. Duplicate Order IDs were observed due to multiple products being associated with the same order, which is expected in transactional datasets.
9. Negative-profit transactions were identified, indicating cases where discounts or operational costs impacted profitability.
10. Data validation confirmed that the dataset is suitable for analysis and supports meaningful business decision-making.